Chapter 4: Small Oscillations

İskender Yalçınkaya, PhD

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In this chapter, we will study how physical systems behave in the vicinity of their stable equilibrium points. Such behavior is often referred to as *small oscillations* because the system exhibits oscillatory motion about its equilibrium position, with a displacement that remains small.

We will first examine the simplest case of oscillatory motion, in which a restoring force acts on the system, pushing it back toward its equilibrium position. We will then consider the effects of a retarding force, which dissipates mechanical energy and causes the amplitude of oscillations to decrease over time. Finally, we will study driven oscillations, in which an external periodic force acts on the system, and investigate how the resulting motion depends on the frequency of the driving force, including the phenomenon of resonance. For simplicity, we will restrict our attention to one-dimensional systems throughout this chapter.

1 Simple Oscillations

We begin by considering the motion of a particle under a conservative force described by a potential energy function $U(x)$. We make the following three assumptions:

(i) $U(x)$ is sufficiently smooth to allow a Taylor expansion about the equilibrium point.

(ii) The potential has a local minimum at $x = x_0$, with $U'(x_0) = 0$ and $U''(x_0) > 0$, so that x_0 is a stable equilibrium point.

(iii) The displacement from equilibrium, $|x - x_0|$, remains "sufficiently small".

Assumption (i) allows us to expand the potential energy about the equilibrium point x_0 using a Taylor series:

$$U(x) = U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2 + \frac{1}{6}U'''(x_0)(x - x_0)^3 + \dots \qquad (1)$$

According to assumption (ii), the first derivative vanishes at equilibrium, $U'(x_0) = 0$, while the second derivative is positive, $U''(x_0) > 0$. Furthermore, assumption (iii) allows us to

neglect the third and higher-order terms compared to the quadratic term. This is what we actually mean when we say that the displacement is "sufficiently small" in assumption (iii). Since the reference level of potential energy can be chosen arbitrarily, we may also set $U(x_0) = 0$ without loss of generality. We therefore obtain the quadratic approximation

$$U(x) \approx \frac{1}{2}k(x - x_0)^2, \tag{2}$$

where $k \equiv U''(x_0) > 0$. It is worth emphasizing that the approximation for the potential energy follows directly from our assumptions and is not limited to specific systems such as springs or pendulums. A wide range of physical systems can be described by a quadratic potential near their stable equilibrium points, provided that the second derivative of the potential at equilibrium is positive. This allows us to study the small oscillations of different physical systems using the same mathematical framework.

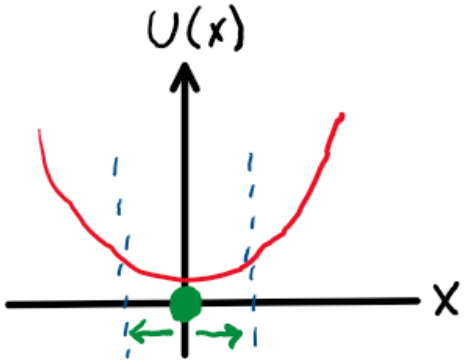


Figure 1: Small oscillations of a particle about a stable equilibrium point under potential energy $U(x)$. The particle oscillates about the equilibrium position x_0 with a small amplitude.

Our initial assumption of having a potential energy function directly means that we are considering a conservative force. Therefore, using the quadratic approximation, we can derive the force acting on the particle:

$$F(x) = -\frac{dU}{dx} \approx -k(x - x_0). \tag{3}$$

This is known as *Hooke's law*. The force is proportional to the displacement from equilibrium and always points toward the equilibrium position x_0. For this reason, it is called a *restoring force*. The constant k is called the *force constant* or *spring constant*, and it characterizes the stiffness of the system. A larger value of k corresponds to a stiffer system, which requires more force to achieve the same displacement from equilibrium.

From the experimental point of view, the force constant can be determined by measuring the force required to displace the particle from its equilibrium position and calculating the slope of the force-displacement curve as shown in Fig. 2. The force constant is a fundamental property of the system and depends on its physical characteristics, such as the material properties and geometry.

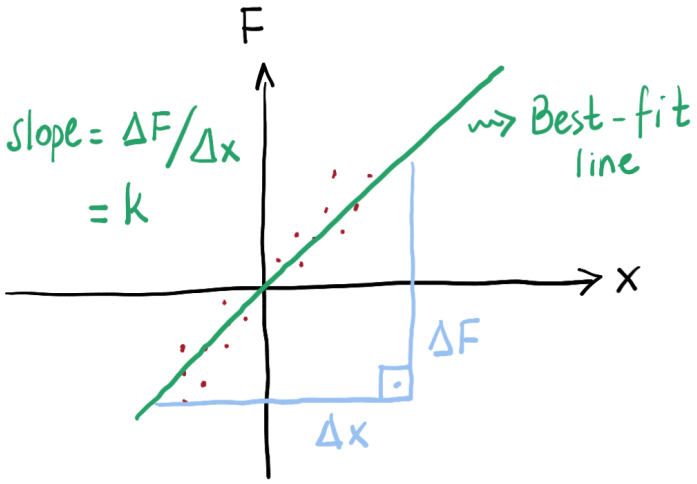


Figure 2: Empirical determination of the force constant k. The slope of the force-displacement curve gives the value of k. Note that the linear best-fit line is a model; the idea is there is always a sufficiently small region around the equilibrium point where the force is linear in the displacement. The data points may scatter according to different models as the displacement increaases.

1.1 Equation of Motion for Simple Oscillations

The dynamics of the particle are governed by Newton's second law:

$$F_{\text{net}} = m\ddot{x}, \tag{4}$$

where m is the mass of the particle. Assuming that the restoring force is the only net force acting on the particle, and choosing the origin at the equilibrium position ($x_0 = 0$), we obtain

$$m\ddot{x} = -kx. \tag{5}$$

Rearranging this equation gives

$$\ddot{x} + \omega_0^2 x = 0, \tag{6}$$

where we have introduced the positive parameter

$$\omega_0 \coloneqq \sqrt{\frac{k}{m}}. \tag{7}$$

This is a second-order linear homogeneous differential equation describing *simple harmonic motion*. The physical meaning of ω_0 will become clear when we solve this equation and examine the resulting motion.

1.2 Solution to the Equation of Motion

We know that the particle oscillates back and forth about the equilibrium position in a periodic way. We saw in Chapter 1 that the exponential function $e^{\lambda t}$ exhibits a wide range

of different types of behavior depending on how the parameter λ is chosen as summarized in the following table:

λ	Behavior of $e^{\lambda t}$
Real and positive ($\lambda > 0$)	Exponential growth
Real and negative ($\lambda < 0$)	Exponential decay
Purely imaginary ($\lambda = i$)	Sinusoidal oscillation
Complex, $\mathrm{Re}(\lambda) < 0$	Oscillation with exponentially decaying amplitude
Complex, $\mathrm{Re}(\lambda) > 0$	Oscillation with exponentially growing amplitude

We know that the solution to the differential equation in Eq. 6 must be oscillatory according to our assumptions. Therefore, it is natural[1] to look for a solution of the form $x(t) = e^{\lambda t}$, where λ will be determined by substituting this ansatz into the differential equation. Differentiating this ansatz twice and pluging it into the equation of motion as given by Eq. 6 gives

$$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0 \implies \lambda^2 + \omega_0^2 = 0. \tag{8}$$

This equation is called the characteristic equation having the two purely imaginary roots

$$\lambda_{1,2} = \pm i\omega_0. \tag{9}$$

This leads to two different solutions $x_{1,2}(t) = e^{\pm i\omega_0 t}$. Not surprisingly, the both solutions have a pure complex factor of the time leading to oscillatory behavior.

A second-order linear homogeneous differential equation requires two initial conditions to determine a unique solution; therefore, the linear combination of two linearly independent solutions, containing two arbitrary constants, is sufficient to describe all possible solutions [2]. Thus, the general solution can be written as

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}, \tag{10}$$

where A and B are arbitrary constants determined by the initial conditions. In general, these constants can be complex. However, since $x(t)$ represents the position of a particle, it must be a real-valued function. To express the solution in a real form, we use Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{11}$$

which also implies

$$e^{-i\theta} = \cos\theta - i\sin\theta. \tag{12}$$

Substituting these expressions into the general solution gives

$$\begin{aligned} x(t) &= A\left[\cos(\omega_0 t) + i\sin(\omega_0 t)\right] + B\left[\cos(\omega_0 t) - i\sin(\omega_0 t)\right] \\ &= (A+B)\cos(\omega_0 t) + i(A-B)\sin(\omega_0 t). \end{aligned} \tag{13}$$

[1] I admit that this is a handwaving argument. For the complete description, we need to study Laplace transforms which is beyond the scope of this text.

[2] The details about the construction of the general solution is a topic of linear algebra and again beyond the scope of this text.

At first glance, the coefficients $A + B$ and $i(A - B)$ may appear to be complex. However, we can show that they must be real by considering the initial conditions. Suppose that the particle has an initial position $x(0) = x_0$ and an initial velocity $\dot{x}(0) = v_0$. Evaluating the position at $t = 0$, we obtain

$$x(0) = A + B = x_0. \tag{14}$$

Differentiating the solution with respect to time gives

$$\dot{x}(t) = -\omega_0(A + B)\sin(\omega_0 t) + i\omega_0(A - B)\cos(\omega_0 t). \tag{15}$$

Evaluating the velocity at $t = 0$, we find

$$\dot{x}(0) = i\omega_0(A - B) = v_0, \tag{16}$$

and therefore

$$A + B = x_0, \qquad i(A - B) = \frac{v_0}{\omega_0}. \tag{17}$$

Since both the initial position x_0 and the initial velocity v_0 are real physical quantities, the combinations $A + B$ and $i(A - B)$ must also be real. Substituting these expressions back into the general solution, we finally obtain

$$x(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0}\sin(\omega_0 t). \tag{18}$$

This expression describes the motion of the particle for any given initial position x_0 and initial velocity v_0. For convenience, we can introduce two real constants, $C = x_0$ and $D = v_0/\omega_0$, and write the general solution in the more compact form

$$x(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t). \tag{19}$$

The constants C and D are determined by the initial conditions.

1.3 Amplitude-Phase Form of the General Solution

This general solution given in Eq. 19 can also be expressed in terms of the *amplitude* and *phase* of the oscillation. To see this, we start from Eq. 19 and use Euler's formula to rewrite the trigonometric functions in terms of complex exponentials:

$$\begin{aligned} x(t) &= C\,\mathrm{Re}\left[e^{i\omega_0 t}\right] + D\,\mathrm{Re}\left[-ie^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Ce^{i\omega_0 t} - iDe^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[(C - iD)e^{i\omega_0 t}\right], \end{aligned} \tag{20}$$

where Re denotes the real part of a complex number. We now introduce the complex number

$$Z := C - iD, \tag{21}$$

which is constant and called a *phasor* or *complex amplitude*. Recall that any nonzero complex number can be expressed in polar form as $Z = Re^{i\phi}$, where $R = |Z|$ is its magnitude and ϕ is its argument. Since $Z = C - iD$, its magnitude and argument can be determined from the real and imaginary parts:

$$R = \sqrt{C^2 + D^2}, \qquad \phi = \arctan\left(-\frac{D}{C}\right). \tag{22}$$

Substituting the polar form of Z into the expression for the displacement gives

$$\begin{aligned} x(t) &= \mathrm{Re}\left[Ze^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i\phi}e^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i(\omega_0 t+\phi)}\right]. \end{aligned} \tag{23}$$

Using Euler's formula, we obtain

$$x(t) = R\cos(\omega_0 t + \phi). \tag{24}$$

This is called the *amplitude-phase form* of the general solution. Recalling that $C = x_0$ and $D = v_0/\omega_0$, the amplitude and the initial phase can be expressed in terms of the initial conditions as

$$R = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}, \qquad \phi = \arctan\left(-\frac{v_0}{\omega_0 x_0}\right). \tag{25}$$

The phasor representation also provides a geometric interpretation of the motion as shown in Fig. 3. The complex quantity $Re^{i(\omega_0 t+\phi)}$ represents a vector of constant length R rotating counterclockwise in the complex plane with angular velocity ω_0. The projection of this rotating vector onto the real axis gives the physical displacement $x(t)$ of the particle.

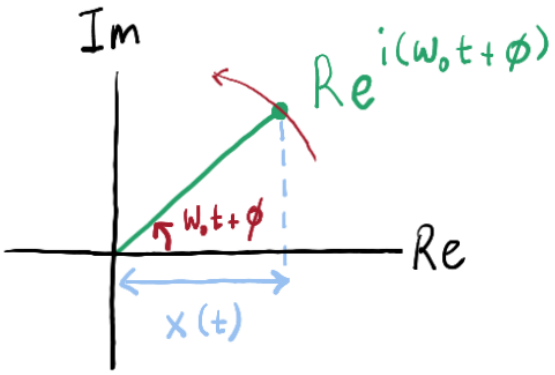


Figure 3: Phasor representation of simple harmonic motion. The complex vector $Re^{i(\omega_0 t+\phi)}$ rotates in the complex plane, and its projection onto the real axis gives the physical displacement $x(t)$.

1.4 Physical Interpretation of the General Solution

In the general solution given in Eq. (24), the constant $R \geq 0$ is called the *amplitude*, representing the maximum displacement of the particle from its equilibrium position. The constant ϕ is called the *initial phase* (or *phase constant*), which determines where in its oscillation cycle the particle is at $t = 0$. In other words, the initial phase determines at what point in the oscillation we start our chronometer. The constant ω_0 is the *angular frequency*, which determines how rapidly the particle oscillates. Recall from Eq. (7) that $\omega_0 = \sqrt{k/m}$, where k is the force constant and m is the mass of the particle. Thus, a stiffer system (larger k) or a lighter particle (smaller m) oscillates more rapidly. In Fig. 4, you can see how $x(t)$ varies with time.

The constant ω_0 is also called the *natural angular frequency* or *natural oscillation frequency* of the system, as it characterizes the intrinsic oscillatory behavior of the system in the absence of any external forces or damping. The natural angular frequency is a fundamental property of the system and depends only on its physical characteristics, such as the mass and stiffness. It is important to note that ω_0 is independent of the initial conditions, meaning that all particles with the same mass and force constant will oscillate with the same natural angular frequency, regardless of their initial position or velocity.

The motion is *periodic*, meaning that both the position and velocity of the particle repeat after a fixed time interval called the *period* T. For simple harmonic motion, the period is

$$T = \frac{2\pi}{\omega_0}. \tag{26}$$

The *frequency* f is the number of complete oscillations per unit time and is given by

$$f = \frac{1}{T} = \frac{\omega_0}{2\pi}. \tag{27}$$

The frequency is measured in hertz (Hz), while the angular frequency is measured in radians per second (rad/s).

As shown in Eq. (25), both the amplitude R and the initial phase ϕ are determined by the initial position and velocity of the particle. Unlike these two quantities, the angular frequency ω_0 depends only on the physical properties of the oscillator and not on its initial conditions. Consequently, the period of simple harmonic motion is independent of the amplitude of oscillation.

Finally, in the special case $R = 0$, the particle remains at its equilibrium position, $x(t) = 0$, and the phase ϕ is undefined.

1.5 Energy of Simple Oscillations

The mechanical energy of a particle undergoing simple oscillation

$$E = K + U = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 \tag{28}$$

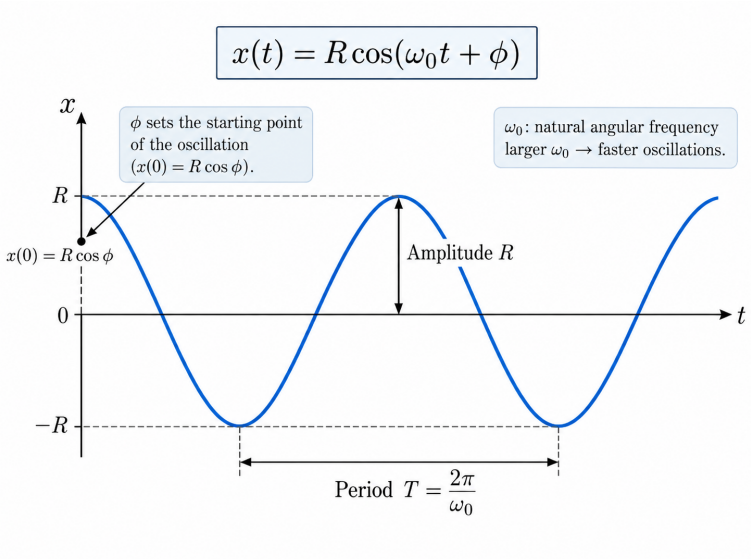


Figure 4: General solution of simple harmonic motion. The particle oscillates about the equilibrium position with amplitude R and angular frequency ω_0. The initial phase ϕ determines the starting point of the oscillation at $t = 0$.

Substituting the general solution $x(t) = R\cos(\omega_0 t + \phi)$ into the expressions for kinetic and potential energies, we find

$$K(t) = \frac{1}{2}m\left[-R\omega_0\sin(\omega_0 t + \phi)\right]^2 = \frac{1}{2}mR^2\omega_0^2\sin^2(\omega_0 t + \phi), \tag{29}$$

$$U(t) = \frac{1}{2}k\left[R\cos(\omega_0 t + \phi)\right]^2 = \frac{1}{2}kR^2\cos^2(\omega_0 t + \phi). \tag{30}$$

Using the relation $k = m\omega_0^2$, we can express the total energy as

$$E = \frac{1}{2}mR^2\omega_0^2\left[\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)\right] = \frac{1}{2}mR^2\omega_0^2. \tag{31}$$

By substituting $\omega_0^2 = k/m$, we can also express the total energy in terms of the force constant k and the amplitude R:

$$E = \frac{1}{2}kR^2. \tag{32}$$

Hence, the mechanical energy of a simple oscillator is constant and proportional to the square of the amplitude. This is a characteristic feature of conservative systems, where energy is conserved and oscillates between kinetic and potential forms without any loss as shown in the Fig. 5, with the maximum kinetic energy occurring when the particle passes through the equilibrium position ($x = 0$) and the maximum potential energy occurring at the turning points of the motion ($x = \pm R$).

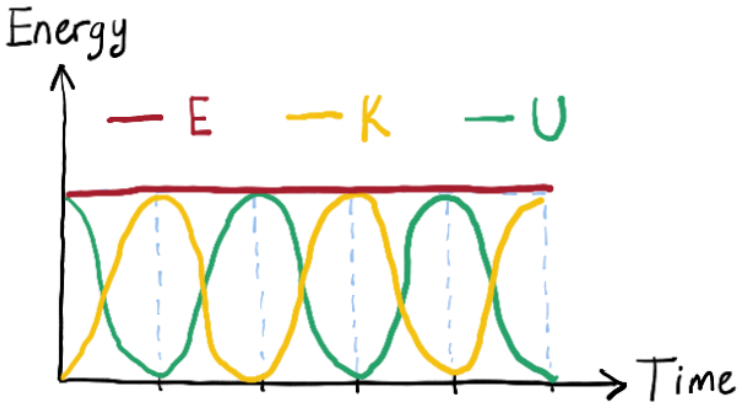


Figure 5: Oscillations of kinetic energy $K(t)$, potential energy $U(t)$, and total energy E for a particle undergoing simple harmonic motion. The total energy remains constant, while the kinetic and potential energies oscillate out of phase with each other.

2 Damped Oscillations

If the oscillating system is not isolated, it may experience a retarding force that dissipates mechanical energy. We will consider a model in which the retarding force is proportional to the velocity of the particle[3], which is a common situation in many physical systems. The retarding force can be expressed as

$$F_{\mathrm{r}} = -b\dot{x}, \tag{33}$$

where b is a positive constant called the *damping coefficient*. The negative sign indicates that the retarding force opposes the motion of the particle. The damping coefficient has units of force per unit velocity (N·s/m) and depends on the properties of the medium through which the particle moves, such as viscosity or air resistance. The total force acting on the particle is then the sum of the restoring force and the retarding force:

$$F_{\mathrm{net}} = F_{\mathrm{restoring}} + F_{\mathrm{retarding}} = -kx - b\dot{x}. \tag{34}$$

This leads to the equation of motion for damped oscillations:

$$m\ddot{x} + b\dot{x} + kx = 0. \tag{35}$$

Rearranging this equation gives

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \tag{36}$$

where we have introduced the positive parameters

$$\gamma \coloneqq \frac{b}{2m}, \qquad \omega_0 \coloneqq \sqrt{\frac{k}{m}}. \tag{37}$$

[3]Note that this is just a simple model that we can easily solve analytically. In real-life situations, the retarding force can be more complex and may depend on other factors such as the shape and size of the particle, the properties of the medium, and the frequency of the oscillation.

We would expect that the damping force would cause the amplitude of oscillations to decrease over time, eventually bringing the particle to rest at its equilibrium position. The parameter γ characterizes the strength of the damping, with larger values corresponding to stronger damping effects. The parameter ω_0 is the natural angular frequency of the undamped system, as we have seen in the previous section.

We know that the exponential function $e^{\lambda t}$ exhibits a wide range of different types of behavior depending on how the parameter λ is chosen. For the damped oscillation equation, we can look for a solution of the form $x(t) = e^{\lambda t}$, as we did it in Section 1.2 for simple oscillations. Plugging this into the differential equation in Eq. 36 leads to the characteristic equation

$$\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0, \tag{38}$$

which has the two roots

$$\lambda_{1,2} = \frac{-2\gamma \pm \sqrt{(2\gamma)^2 - 4\omega_0^2}}{2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{39}$$

The roots can be real or complex, depending on the relative magnitudes of γ and ω_0, leading to three different cases: underdamped ($\gamma < \omega_0$), critically damped ($\gamma = \omega_0$), and overdamped ($\gamma > \omega_0$). Each case corresponds to a different type of motion, which we will analyze in detail in the following subsections.

2.1 Underdamped Oscillations ($\gamma < \omega_0$)

In the underdamped case, the damping is weak enough that the system still oscillates, but with a gradually decreasing amplitude. The roots of the characteristic equation are complex conjugates:

$$\lambda_{1,2} = -\gamma \pm i\omega_1, \tag{40}$$

where we have defined the *damped angular frequency*

$$\omega_1 \coloneqq \sqrt{\omega_0^2 - \gamma^2}. \tag{41}$$

The solution to the differential equation in this case is

$$x(t) = e^{-\gamma t}(Ae^{i\omega_1 t} + Be^{-i\omega_1 t}), \tag{42}$$

where A and B are constants determined by the initial conditions. According to Eq. 24, we can also express the solution in amplitude-phase form as

$$x(t) = (Re^{-\gamma t})\cos(\omega_1 t + \phi), \tag{43}$$

where R is the initial amplitude and ϕ is the initial phase. What we see inside the parantheses is the amplitude of the oscillation, which decays exponentially over time due to the damping effect. On the other hand, the cosine term describes the oscillatory motion with angular

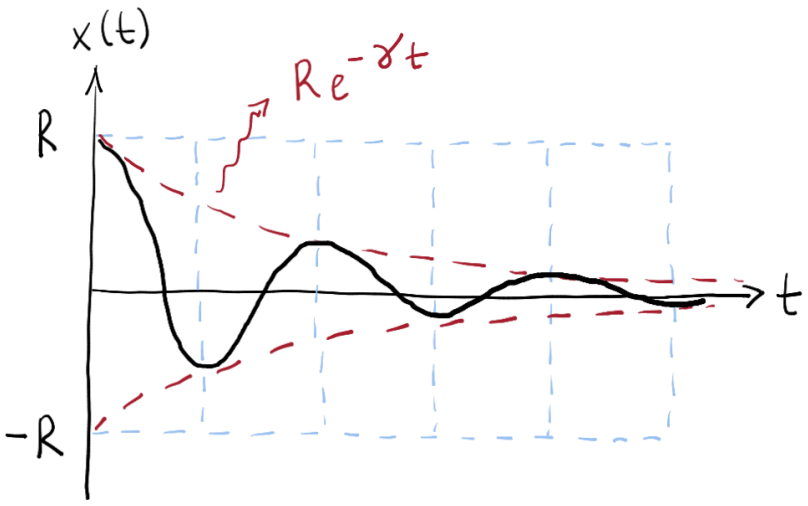


Figure 6: Underdamped oscillations of a particle. The amplitude of oscillation decays exponentially over time due to the damping effect, while the particle continues to oscillate with a damped angular frequency ω_1.

frequency ω_1. Therefore, the motion is a combination of exponential decay and sinusoidal oscillation as shown in Fig. 6.

Directly followed from the definition of ω_1, the damped angular frequency ω_1 is slightly less than the natural angular frequency ω_0, indicating that the oscillations occur at a lower frequency than in the undamped case

$$\omega_1 = \sqrt{\omega_0^2 - \gamma^2} < \omega_0.$$

Strictly speaking, the word "frequency" attributed to periodic motion. Therefore, calling ω_1 a frequency is not strictly correct, since the motion is not strictly periodic due to the damping. However, it is common practice to refer to ω_1 as the damped angular frequency, as it characterizes the rate of oscillation in the presence of damping for a given time interval.

We know that the mechanical energy of a simple oscillator is constant and proportional to the square of its amplitude. In the case of underdamped oscillations, the amplitude decreases exponentially over time, resulting in a corresponding loss of mechanical energy. Since both the displacement and velocity acquire the same exponential factor $e^{-\gamma T}$ after one complete period T, the mechanical energy satisfies

$$E(t+T) = E(t)e^{-2\gamma T}. \tag{44}$$

Consequently, after n complete periods,

$$E(nT) = E_0 e^{-2\gamma nT}, \tag{45}$$

where E_0 is the initial mechanical energy. The decrease in mechanical energy is due to dissipation, typically involving its conversion into heat or other forms of energy.

2.2 Critically Damped Oscillations ($\gamma = \omega_0$)

In the critically damped case, the damping is just strong enough to prevent oscillations. The roots of the characteristic equation are real and equal:

$$\lambda_{1,2} = -\gamma. \tag{46}$$

Therefore, we do not have two distinct solutions. We mentioned in Sec. 1.2 that the general solution for a second-order linear differential equation with constant coefficients is a linear combination of two independent solutions. In this case, we have only one independent solution, which is $e^{-\gamma t}$. To find a second independent solution, we return to the equation of motion, which, for $\gamma = \omega_0$, takes the form

$$\ddot{x} + 2\gamma\dot{x} + \gamma^2 x = 0. \tag{47}$$

Multiplying both sides by $e^{\gamma t}$, we obtain

$$e^{\gamma t}\ddot{x} + 2\gamma e^{\gamma t}\dot{x} + \gamma^2 e^{\gamma t}x = 0. \tag{48}$$

Note that the left-hand side is precisely the second derivative of the product $e^{\gamma t}x(t)$, as can be verified by applying the product rule twice:

$$\frac{d^2}{dt^2}\left[e^{\gamma t}x(t)\right] = e^{\gamma t}\ddot{x} + 2\gamma e^{\gamma t}\dot{x} + \gamma^2 e^{\gamma t}x. \tag{49}$$

Therefore, the equation of motion can be rewritten as

$$\frac{d^2}{dt^2}\left[e^{\gamma t}x(t)\right] = 0. \tag{50}$$

Integrating twice with respect to time gives

$$e^{\gamma t}x(t) = A + Bt, \tag{51}$$

where A and B are arbitrary constants. Solving for $x(t)$, we obtain

$$x(t) = (A + Bt)e^{-\gamma t}. \tag{52}$$

We can now identify the two linearly independent solutions:

$$x_1(t) = e^{-\gamma t}, \qquad x_2(t) = te^{-\gamma t}. \tag{53}$$

Indeed, neither solution can be expressed as a constant multiple of the other, so they are linearly independent. The constants A and B can be determined from the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$. Substituting these conditions gives

$$A = x_0, \qquad B = v_0 + \gamma x_0. \tag{54}$$

Thus, the motion of a critically damped oscillator is described by

$$x(t) = \left[x_0 + (v_0 + \gamma x_0)t\right] e^{-\gamma t}. \tag{55}$$

Since both $e^{-\gamma t}$ and $te^{-\gamma t}$ approach zero as $t \to \infty$, the particle eventually returns to its equilibrium position without undergoing oscillations. Depending on the initial conditions, the particle may cross the equilibrium position once, but it cannot oscillate back and forth around it.

2.3 Overdamped Motion ($\gamma > \omega_0$)

In the overdamped case, the damping is sufficiently strong to prevent oscillations. The roots of the characteristic equation are real and distinct:

$$\lambda_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{56}$$

Since $\gamma > \omega_0$, both roots are negative, with $\lambda_2 < \lambda_1 < 0$. The general solution to the differential equation is

$$x(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}, \tag{57}$$

where A and B are constants determined by the initial conditions.

Since both exponential terms decay with time, the particle eventually approaches its equilibrium position. However, the two terms decay at different rates. Since $\lambda_1 > \lambda_2$, the term $e^{\lambda_1 t}$ decays more slowly and dominates the motion at sufficiently long times, provided $A \neq 0$.

Unlike the critically damped case, overdamping generally results in a slower approach to equilibrium. Although the particle may cross the equilibrium position once, depending on its initial conditions, it cannot undergo oscillatory motion about equilibrium.

Since $\lambda_2 < \lambda_1 < 0$, both exponential terms decay with time. At sufficiently long times, the term $e^{\lambda_1 t}$ dominates, provided $A \neq 0$. Notice that

$$\lambda_1 = -\frac{\omega_0^2}{\gamma + \sqrt{\gamma^2 - \omega_0^2}} > -\omega_0, \tag{58}$$

where we have used $\gamma > \omega_0$. Thus, the dominant exponential decays more slowly than the exponential factor $e^{-\omega_0 t}$ appearing in the critically damped solution. Overdamping therefore generally produces a slower relaxation toward equilibrium.

Furthermore, the particle can cross equilibrium only when $x(t) = 0$, which, for nonzero A and B, requires

$$e^{(\lambda_1 - \lambda_2)t} = -\frac{B}{A}. \tag{59}$$

Since the left-hand side is a strictly increasing function of time, this equation can have at most one solution. Consequently, the particle may cross equilibrium once, depending on its initial conditions, but it cannot undergo oscillatory motion.

2.4 Comparison of Damping Regimes

The underdamped motion is the only case that exhibits oscillatory behavior, while the critically damped and overdamped cases do not oscillate. The critically damped case represents the fastest approach to equilibrium without oscillation, while the overdamped case results in a slower approach to equilibrium. The underdamped case exhibits oscillatory motion with exponentially decaying amplitude, while the critically damped and overdamped cases do not

oscillate and approach equilibrium at different rates. The comparison of the three damping regimes is illustrated in Fig. 7.

The critically damped case is often considered the optimal damping scenario for systems that require a quick return to equilibrium without overshooting, such as in automotive suspension systems, certain types of mechanical clocks, pneumatic-actuated systems and screen door closure systems. The overdamped case, while slower, may be preferred in situations where overshooting must be avoided at all costs, even if it means a longer time to reach equilibrium.

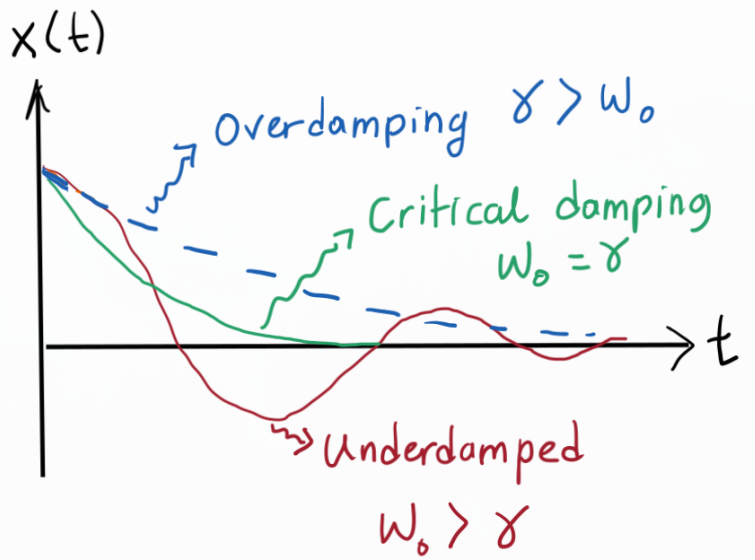


Figure 7: Comparison of the three damping regimes: underdamped, critically damped, and overdamped. The underdamped case exhibits oscillatory motion with exponentially decaying amplitude, while the critically damped and overdamped cases do not oscillate and approach equilibrium at different rates.

2.5 Quality Factor

The *quality factor* Q is a dimensionless quantity that characterizes how weakly an oscillating system is damped. We know that an undamped oscillator conserves its mechanical energy, whereas a damped oscillator gradually loses energy. A natural way to characterize the strength of damping is therefore to compare the mechanical energy stored in the oscillator with the energy dissipated during one complete oscillation.

For weak damping ($\gamma \ll \omega_0$), the quality factor can be expressed approximately as

$$Q \approx 2\pi \frac{E(t)}{\Delta E}, \tag{60}$$

where $E(t)$ is the mechanical energy at tan arbitrary time and ΔE is the energy dissipated over the subsequent oscillation period T. The factor 2π is included by convention. A high quality factor corresponds to weak damping, meaning that the oscillator loses only a small

fraction of its mechanical energy during each cycle. To relate the quality factor to the damping parameter γ, we recall that the displacement and velocity of an underdamped oscillator is given by

$$x(t) = R_0 e^{-\gamma t}\cos(\omega_1 t + \phi), \qquad \dot{x}(t) = -R_0 e^{-\gamma t}\left[\gamma\cos(\omega_1 t + \phi) + \omega_1 \sin(\omega_1 t + \phi)\right]. \tag{61}$$

After one complete oscillation period $T = 2\pi/\omega_1$, the trigonometric functions return to their original values. Consequently, both the displacement and velocity are multiplied by the same exponential factor:

$$x(t+T) = e^{-\gamma T}x(t), \qquad v(t+T) = e^{-\gamma T}v(t). \tag{62}$$

Since the total mechanical energy is given by $E(t) = \frac{1}{2}mv^2(t) + \frac{1}{2}kx^2(t)$, we obtain

$$\begin{aligned} E(t+T) &= \frac{1}{2}mv^2(t+T) + \frac{1}{2}kx^2(t+T) \\ &= e^{-2\gamma T}\left[\frac{1}{2}mv^2(t) + \frac{1}{2}kx^2(t)\right]. \end{aligned} \tag{63}$$

Therefore,

$$E(t+T) = E(t)e^{-2\gamma T}. \tag{64}$$

The mechanical energy dissipated during one complete oscillation is

$$\begin{aligned} \Delta E &= E(t) - E(t+T) \\ &= E(t)\left(1 - e^{-2\gamma T}\right). \end{aligned} \tag{65}$$

For weak damping ($\gamma \ll \omega_0$), we have $2\gamma T \ll 1$. Using the first-order Taylor approximation

$$e^{-2\gamma T} \approx 1 - 2\gamma T, \tag{66}$$

we obtain

$$\frac{\Delta E}{E(t)} \approx 2\gamma T. \tag{67}$$

Substituting this result into the expression for the quality factor gives

$$Q \approx \frac{2\pi}{2\gamma T} = \frac{\pi}{\gamma T} = \frac{\omega_1}{2\gamma}. \tag{68}$$

Since $\omega_1 = \sqrt{\omega_0^2 - \gamma^2} \approx \omega_0$ for weak damping, we obtain

$$Q \approx \frac{\omega_0}{2\gamma}. \tag{69}$$

For the linearly damped harmonic oscillator, this last expression is conventionally adopted as the exact definition of the quality factor:

$$Q \coloneqq \frac{\omega_0}{2\gamma}. \tag{70}$$

Its interpretation in terms of the energy dissipated per oscillation, however, is valid approximately in the weak-damping regime as described above. Rearranging the energy-based expression for the quality factor gives

$$\frac{\Delta E}{E(t)} \approx \frac{2\pi}{Q}. \tag{71}$$

This relation provides a direct physical interpretation of Q: the larger the quality factor, the smaller the fraction of mechanical energy dissipated during each oscillation. For example, an oscillator with $Q = 100$ loses approximately

$$\frac{\Delta E}{E(t)} \approx \frac{2\pi}{100} \approx 0.063, \tag{72}$$

or about 6% of its mechanical energy per cycle.The quality factor also allows us to express the damped angular frequency ω_1 in terms of the natural angular frequency ω_0. Using Eq. (70), we find

$$\omega_1 = \sqrt{\omega_0^2 - \gamma^2} = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}. \tag{73}$$

For example, when $Q = 10$, we obtain $\omega_1 \approx 0.9987\omega_0$. Thus, for weak damping, the oscillation frequency remains very close to the natural frequency, even though the amplitude gradually decreases with time. Even for $Q = 2$, we have $\omega_1 \approx 0.968\omega_0$, which is still a relatively small deviation from the natural frequency. To obtain another physical interpretation of Q, let us determine how many oscillation cycles occur before the amplitude decreases to $1/e$ of its initial value. The amplitude decays according to

$$R(t) = R_0 e^{-\gamma t}, \tag{74}$$

where R_0 is the initial amplitude. The time required for the amplitude to decrease to R_0/e is therefore

$$t_{1/e} = \frac{1}{\gamma}. \tag{75}$$

The number of oscillation cycles during this time is

$$N = \frac{t_{1/e}}{T} = \frac{\omega_1}{2\pi\gamma}. \tag{76}$$

For weak damping, $\omega_1 \approx \omega_0$, and hence

$$N \approx \frac{Q}{\pi} \implies Q \approx \pi N. \tag{77}$$

This result shows that the quality factor is directly proportional to the number of oscillation cycles required for the amplitude to decrease to $1/e$ of its initial value. For example, an oscillator with $Q = 100$ completes approximately 32 cycles during this time. Thus, the quality factor provides two complementary physical interpretations of damping: it characterizes both the fraction of mechanical energy dissipated during each oscillation cycle and the number of cycles over which the oscillation amplitude significantly decreases.

3 Phase Space and Phase Trajectories

The *state* of a particle at any instant is defined to be specified by its position $x(t)$ and velocity $\dot{x}(t)$. These quantities can be regarded as coordinates in a two-dimensional space called *phase space*, with position along the horizontal axis and velocity along the vertical axis. The state of the particle is then represented by a vector in phase space,

$$\mathbf{X}(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix}. \tag{78}$$

and as time progresses, this vector traces out a curve called a *phase trajectory* as shown in Fig. 8(a). The phase trajectory provides a visual representation of the motion of the particle, showing how its position and velocity change over time.

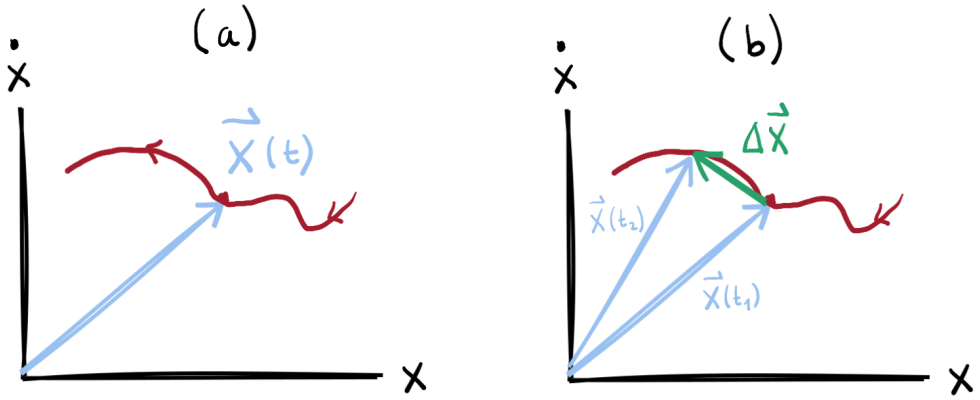


Figure 8: (a) State of a mechanical system in phase space. The state of the particle is represented by a vector $\mathbf{X}(t)$, which traces out a phase trajectory as time progresses. (b) Displacement of the state vector in phase space defining the "velocity" of the state vector, $V_{\text{avg}} = \frac{\Delta \mathbf{X}}{\Delta t}$.

Note that this vector lives in the abstract phase space, which is a two-dimensional space with coordinates $(x, \dot{x})$. It does not correspond to a point in physical space. However, it is still a vector in the sense that it has both magnitude and direction. For example, if we consider the time rate of change of this vector, we consider the time rate of change of the each component as shown in Fig. 8(b). Thus, we get the velocity of the particle in phase space,

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix}. \tag{79}$$

where the first component is the velocity and the second component is the acceleration of the particle. The second component is defined by Newton's second law.

3.1 Time Evolution in Phase Space for Oscillatory Motion

Eliminating time from the equations for position and velocity for small oscillations leads to the equation of the phase trajectory:

$$x(t) = R\cos(\omega_0 t + \phi), \tag{80}$$
$$\dot{x}(t) = -R\omega_0 \sin(\omega_0 t + \phi). \tag{81}$$

Squaring both equations, adding them, and using the identity $\sin^2\theta + \cos^2\theta = 1$, we obtain

$$\frac{x^2}{R^2} + \frac{\dot{x}^2}{R^2\omega_0^2} = 1. \tag{82}$$

This is the equation of an ellipse in phase space, with semi-axes R and $R\omega_0$. Recall that the total mechanical energy of a simple harmonic oscillator is conserved and can be expressed in terms of its amplitude as

$$E = \frac{1}{2}kR^2 = \frac{1}{2}m\omega_0^2 R^2 \implies R^2 = \frac{2E}{k} = \frac{2E}{m\omega_0^2}. \tag{83}$$

Substituting this expression into the equation of the ellipse gives

$$\frac{x^2}{2E/k} + \frac{\dot{x}^2}{2E/m} = 1. \tag{84}$$

Thus, semi-major and semi-minor axes of the ellipse are determined by the total mechanical energy E of the oscillator as shown in Fig. 9(a). The phase trajectories form a family of ellipses, each corresponding to a different value of the conserved mechanical energy E as shown in Fig. 9(b). As the particle oscillates, its state moves clockwise along the corresponding ellipse, completing one revolution during each oscillation period as shown in Fig. 8. The time evolution of the state vector is governed by the equation

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix}\dot{x}(t)\\ \ddot{x}(t)\end{bmatrix} = \begin{bmatrix}\dot{x}(t)\\ -\omega_0^2 x(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\omega_0^2 & 0\end{bmatrix}\begin{bmatrix}x(t)\\ \dot{x}(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\omega_0^2 & 0\end{bmatrix}\mathbf{X}(t). \tag{85}$$

where the matrix

$$\hat{S} \coloneqq \begin{pmatrix}0 & 1\\ -\omega_0^2 & 0\end{pmatrix} \tag{86}$$

is called the *system matrix*. By using this terminology, now we can write the time evolution of the sistem in a compact form as

$$\dot{\mathbf{X}} = \hat{S}\mathbf{X}. \tag{87}$$

This is a first-order linear differential equation for the state vector $\mathbf{X}(t)$, which can be solved using standard techniques which is out of the scope of this text. However, even without getting into the details of the solution of this equation, a lot of important conclusion can be drawn from this simple relation as listed below:

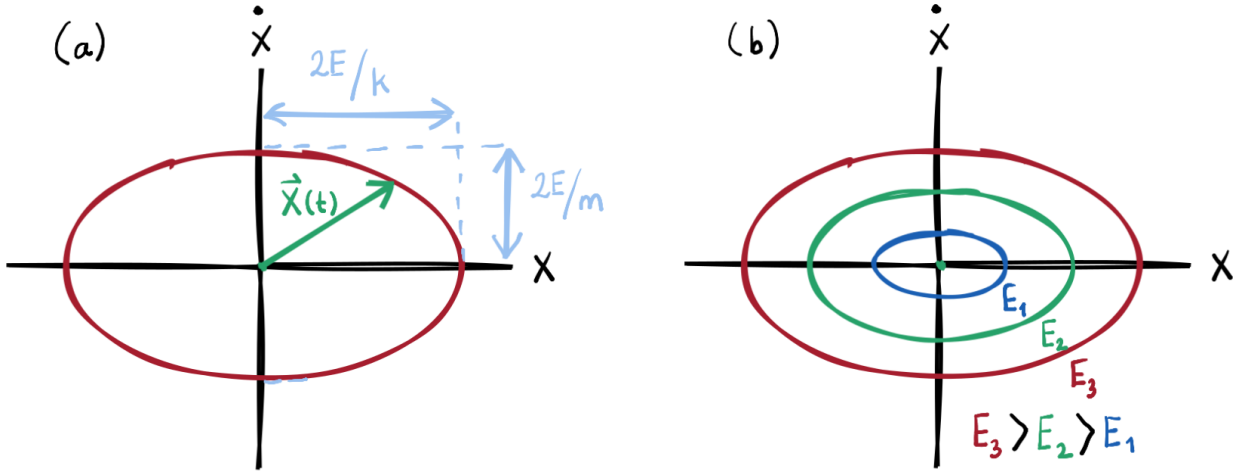


Figure 9: (a) Phase trajectory of a simple harmonic oscillator in phase space. (b) The trajectory is an ellipse, with semi-major and semi-minor axes determined by the total mechanical energy E of the oscillator. Note that the phase trajectories bolong to different energies never intersect each other. If they did, a given initial state would lead to two different energies, which is not possible.

Conclusion 1: The time rate of change of a vectoral quantity (in this case, the state vector $\mathbf{X}$) can be obtained by reorganizing its components according to the prescription given by the system matrix $\hat{S}$:

$$\frac{d}{dt}\begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix}\begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix}. \tag{88}$$

Conclusion 2: Note that the system matrix is only a function of the natural angular frequency ω_0 of the system. Therefore, the time evolution of the state vector is determined solely by the properties of the system, and not by the initial conditions. The initial conditions only determine which phase trajectory the system will follow in phase space.

Conclusion 3: Once we are given the initial conditions at some arbitrary time t_0, $x(t_0)$ and $\dot{x}(t_0)$, or in other words, if we are given the initial "state" $\mathbf{X}(t_0)$ of the system, we can approximately determine the state of the system at a later time $t_0 + dt$

$$\mathbf{X}(t_0 + dt) \approx \mathbf{X}(t_0) + \dot{\mathbf{X}}(t_0)dt = \mathbf{X}(t_0) + \hat{S}\mathbf{X}(t_0)dt. \tag{89}$$

The quality of the approximation improves as dt becomes smaller. By repeating this process, we can determine the state of the system at any later time $t_0 + ndt$ for any positive integer n. In this way, we can trace out the phase trajectory of the system in phase space.

Note that the state of the system at the later time only depends on the current state of the system and the system properties, and not on the history of the system. This is a manifestation of the *Markov property*, which states that the future state of a system depends only on its present state, and not on its past states. This property is a fundamental feature of many physical systems, and it allows us to describe their dynamics using differential equations.

Conclusion 4: Note again that the system matrix $\hat{S}$ is a "passive transformation" such that it does not change in time once the system properties are set, and it acts on any initial state vector in the phase space in the same way. This leads us to conceptually generalize the idea of time evolution in such a way that we can start thinking about the phase space consisting of infinitely many number of possible initial states represented by dots so that we can assign a "velocity vector" associated with the transformation $\hat{S}$ to each of this dots converting this scalar field into a vector field in the phase space as shown in Fig. 10. This velocity vector field does not change in time. Therefore, if we pictoriolly represent initial state of the system by a "ball" in the phase space, then the matrix $\hat{S}$ define the local "kickers" for each point in the phase space so that whenever they have a ball in front of them, they kick it in the predefined (by the system matrix) certain direction with a certain speed and pass it to the one next to it. In this way, the ball will move in the phase space and trace out a phase trajectory as time progresses. This is a very useful way of visualizing the dynamics of the system in phase space, and it can be used to gain insight into the behavior of the system under different initial conditions.

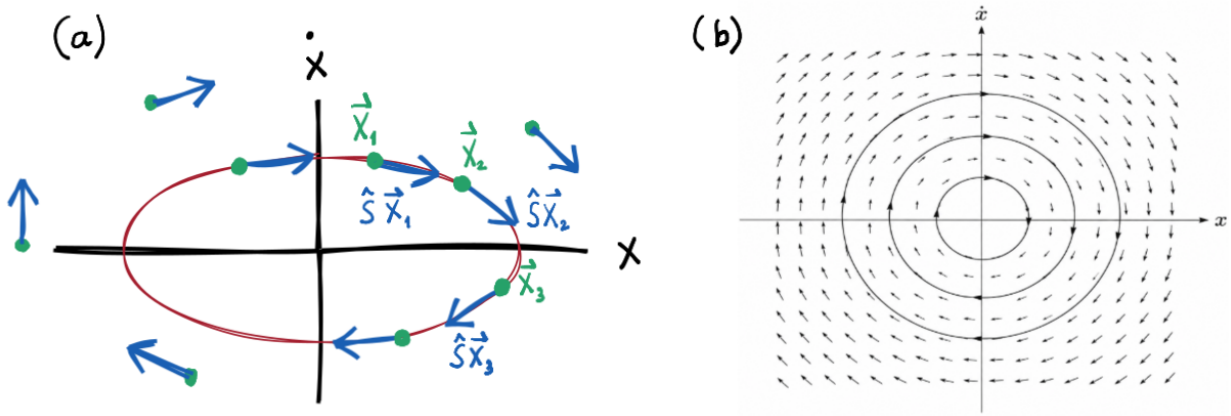


Figure 10: Phase space vector field. (a) Starting from an initial state, the state vector traces out a phase trajectory as it moves through the vector field. (b) The system matrix $\hat{S}$ defines a velocity vector at each point in phase space, which determines the direction and speed of the state vector's motion.

For the damped oscillation, time rate of change of the state vector becomes

$$\frac{d\mathbf{X}}{dt} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}(t) \\ -2\gamma\dot{x} - \omega_0^2 x(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{bmatrix} \mathbf{X}(t). \tag{90}$$

TO BE CONTINUED...

4 Driven Oscillations

We will consider now that the particle which does damped oscillatory motion is also subjected to an external periodic driving force, $F(t)$. These kind of oscillations are called *driven*

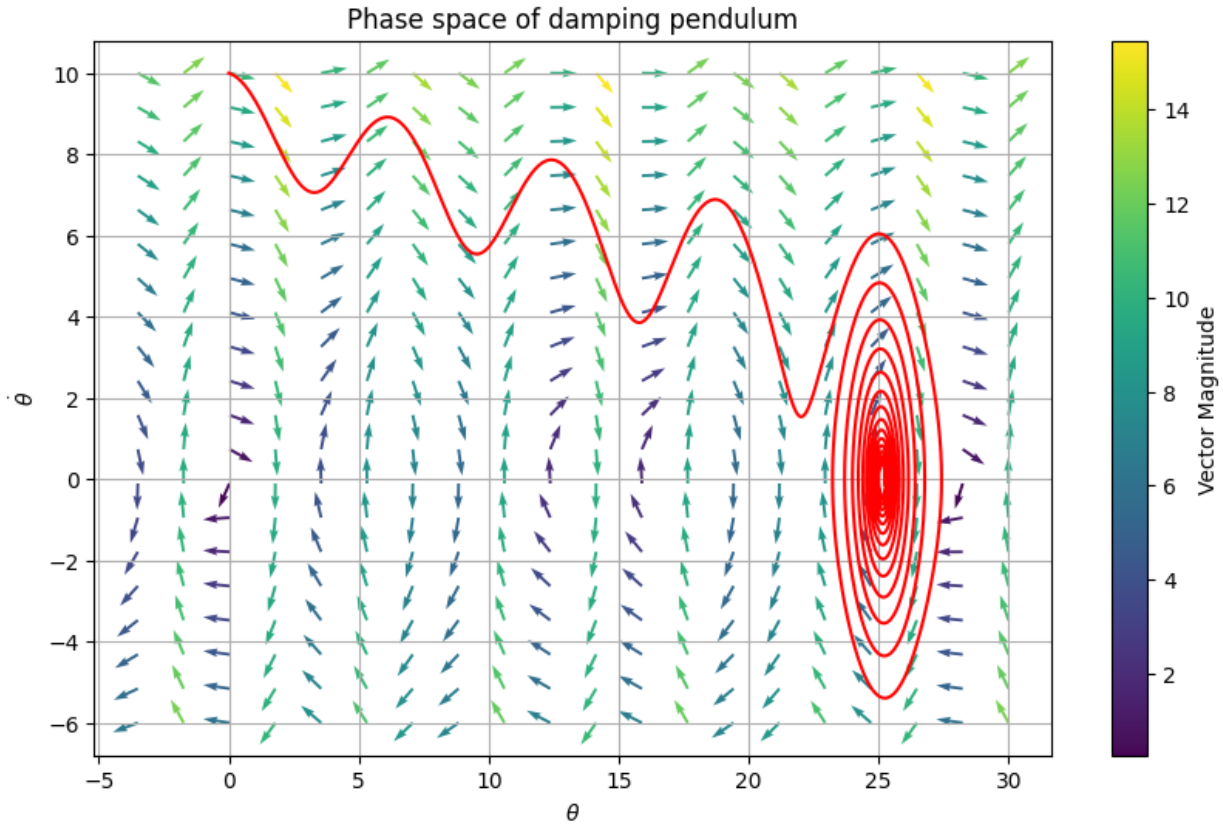


Figure 11: Phase trajectory of a damped harmonic oscillator in phase space. The trajectory is a spiral, with the particle eventually approaching the equilibrium position at the origin.

oscillations.

An intuitive example of driven oscillations is a child on a swing. The child can be thought of as a damped oscillator, with the swing providing the restoring force. If the child pumps their legs at the right frequency, they can increase the amplitude of their swinging motion. In this case, the external driving force is provided by the child's legs, which are applying a periodic force to the swing.

The net force acting on the particle is the sum of the restoring force, the damping force, and the external driving force:

1. Restoring force: $F_{\text{restoring}} = -kx(t)$
2. Damping force: $F_{\text{damping}} = -b\dot{x}(t)$
3. External driving force: $F_{\text{driving}} = F(t) = F_0 \cos(\omega t)$

In general, the external driving force can be any function of time, but for simplicity, we will consider a sinusoidal driving force of the form $F(t) = F_0 \cos(\omega t)$, where F_0 is the amplitude of the driving force and ω is its angular frequency. The total force acting on the system and

the corresponding equation of motion can be written as

$$F_{\text{net}} = F_{\text{r}} + F_{\text{d}} + F_{\text{drive}} = -kx(t) - b\dot{x}(t) + F_0 \cos(\omega t) = m\ddot{x}(t). \tag{91}$$

Rearranging this equation gives the equation of motion for a driven damped harmonic oscillator:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = F_0 \cos(\omega t). \tag{92}$$

Or, in terms of the natural angular frequency ω_0 and the damping constant γ, we can write

$$\ddot{x}(t) + 2\gamma\dot{x}(t) + \omega_0^2 x(t) = \frac{F_0}{m} \cos(\omega t). \tag{93}$$

This is a second-order linear nonhomogeneous differential equation with constant coefficients. It is nonhomogeneous because of the explicit time-dependent driving force on the right-hand side. Before getting into the solution of this equation, let us first discuss what would we expect from the solution.

Assume a weakly damped system was oscillating on its own approximately with frequency ω_0. If an external force starts "shaking" the system, in the beginning, the system will resist to obey the external force because of its intertia and previous motion. However, as the external force persistently continues to act on the system, the system will eventually "give in" and start to oscillate with the same frequency as the external force. This is called *steady-state motion*. In addition to this steady-state motion, there will also be a transient motion that depends on the initial conditions of the system. The transient motion will eventually die out due to damping, leaving only the steady-state motion.

From this point of view, we would expect two different behaviors from the general solution of the equation of motion: a transient motion that decays over time and a steady-state motion that persists indefinitely. The transient motion is called the *complementary solution* $x_{\text{c}}(t)$ and it is the same as the solution of the homogeneous equation (the equation without the driving force). The steady-state motion is called the *particular solution* $x_{\text{p}}(t)$ and it is a specific solution of the nonhomogeneous equation that depends on the form of the driving force. Therefore, the general solution of the equation of motion can be expressed as the sum of the complementary and particular solutions:

$$x(t) = x_{\text{c}}(t) + x_{\text{p}}(t). \tag{94}$$

We already know from previous section that the complementary solution dies out as $t \to \infty$:

$$\lim_{t\to\infty} x_{\text{c}}(t) = 0 \implies \lim_{t\to\infty} x(t) = x_{\text{p}}(t). \tag{95}$$

Two examples of this combined behavior of particular and complementary solutions is given in Fig 12. Following the above discussion, we actually know the form of the particular solution. Since the driving force is sinusoidal, we expect the particular solution to also be sinusoidal with the same frequency as the driving force, but with a different amplitude and phase. Therefore, we can write

$$x_{\text{p}}(t) = A \cos(\omega t - \delta), \tag{96}$$

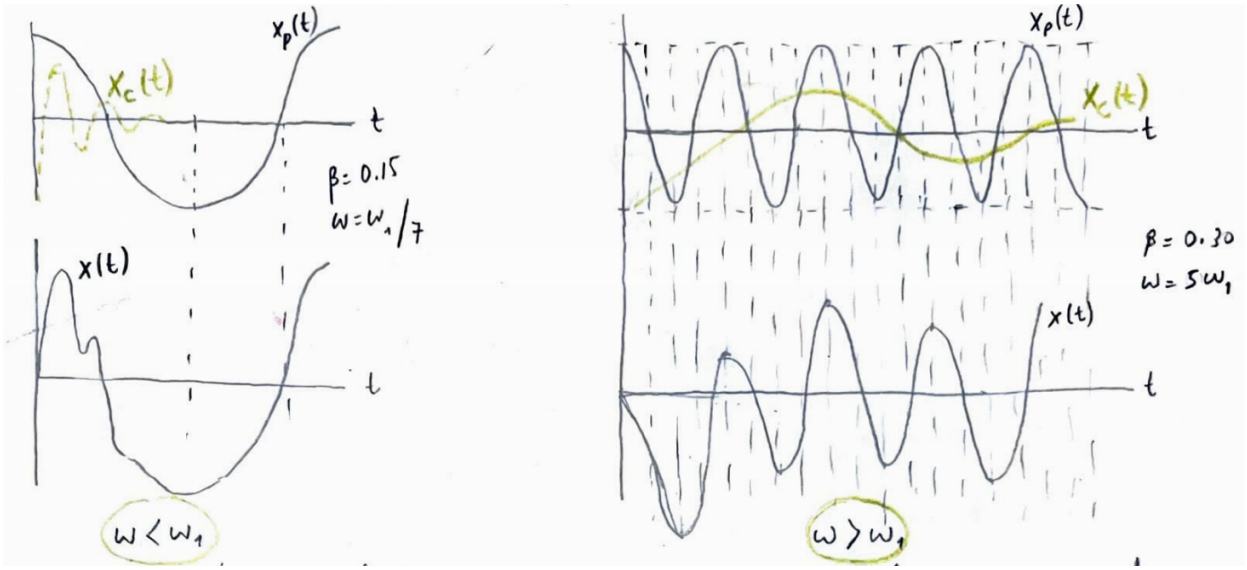


Figure 12: Two examples of driven damped harmonic oscillators with different driving frequencies. The transient motion dies out over time, leaving only the steady-state motion.

where A is the amplitude of the steady-state motion and δ is the phase difference between the driving force and the steady-state motion. This form of the solution provides us with the most general description of the steady-state motion of a driven damped harmonic oscillator regarding the points discussed above. The amplitude A and phase δ can be determined by substituting the particular solution into the equation of motion and solving for A and δ. This can be done by differentiating the particular solution twice and substituting it into the equation of motion. However, the calculations can be quite involved because of the trigonometric functions. Instead, we can so the same calculations in a more elegant way by using complex numbers.

Instead of using Eq. (96) as a solution ansatz, let us switch to the complex plane by expanding it with a complex component:

$$z_{\mathrm{p}}(t) = A\cos(\omega t - \delta) + iA\sin(\omega t - \delta) = Ae^{i(\omega t - \delta)}. \tag{97}$$

From now on, we also think that the equation of motion is also extended to the complex plane by allowing the position $x(t)$ to be a complex function $z(t)$. Therefore, we can write the equation of motion (Eq. (93)) in the complex plane as

$$\ddot{z}(t) + 2\gamma\dot{z}(t) + \omega_0^2 z(t) = \frac{F_0}{m}e^{i\omega t}. \tag{98}$$

Note that we effectively changed nothing by switching to the complex plane since derivation is a linear operation, and actually, it corresponds to considering the same differential equation twice, once for the real part and once for the imaginary part,

$$\frac{d^2}{dt^2}\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] + 2\gamma\frac{d}{dt}\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] + \omega_0^2\left[x_{\mathrm{p}}(t) + iy_{\mathrm{p}}(t)\right] = \frac{F_0}{m}\left[\cos(\omega t) + i\sin(\omega t)\right]. \tag{99}$$

where $x_{\mathrm{p}}(t)$ and $y_{\mathrm{p}}(t)$ are the real and imaginary parts of the complex solution $z_{\mathrm{p}}(t)$, respectively. As you can see real and imaginary part do not mix together. The real part of the complex solution corresponds to the physical solution we are interested in, while the imaginary part is an auxiliary function that helps us simplify the calculations. The rest of the calculation is straightforward. We can differentiate the complex solution twice and substitute it into the equation of motion to solve for the amplitude A and phase δ:

$$\dot{z}_{\mathrm{p}}(t) = i\omega A e^{i(\omega t - \delta)}, \tag{100}$$

$$\ddot{z}_{\mathrm{p}}(t) = -\omega^2 A e^{i(\omega t - \delta)}. \tag{101}$$

Substituting these expressions into the equation of motion gives

$$\left[-\omega^2 + 2i\gamma\omega + \omega_0^2\right] A e^{i(\omega t - \delta)} = \frac{F_0}{m} e^{i\omega t}. \tag{102}$$

Dividing both sides by $e^{i\omega t}$ and rearranging gives

$$\left[-\omega^2 + 2i\gamma\omega + \omega_0^2\right] A = \frac{F_0}{m} e^{i\delta} = \frac{F_0}{m}\left[\cos(\delta) + i\sin(\delta)\right]. \tag{103}$$

Real and imaginary parts of this equation can be separated to give two equations for the amplitude A and phase δ:

$$A\left(\omega_0^2 - \omega^2\right) = \frac{F_0}{m}\cos(\delta), \tag{104}$$

$$2\gamma\omega A = \frac{F_0}{m}\sin(\delta). \tag{105}$$

Squaring both equations and adding them gives

$$A^2\left[\left(\omega_0^2 - \omega^2\right)^2 + (2\gamma\omega)^2\right] = \left(\frac{F_0}{m}\right)^2. \tag{106}$$

Taking the square root of both sides gives the amplitude of the steady-state motion:

$$A = \frac{F_0/m}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + (2\gamma\omega)^2}}. \tag{107}$$

The phase difference δ can be determined by taking the ratio of the imaginary and real parts of the equation:

$$\tan(\delta) = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \implies \delta = \arctan\left(\frac{2\gamma\omega}{\omega_0^2 - \omega^2}\right). \tag{108}$$

The amplitude and phase equations are the most important results of this section, and they provide a complete description of the steady-state motion of a driven damped harmonic oscillator. The amplitude equation shows how the amplitude of the steady-state motion depends on the driving frequency ω, the natural frequency ω_0, the damping constant γ, and the amplitude of the driving force F_0. The phase equation shows how the phase difference between the driving force and the steady-state motion depends on the same parameters. In Fig. 13, we plot the amplitude and phase equations as functions of the driving frequency ω.

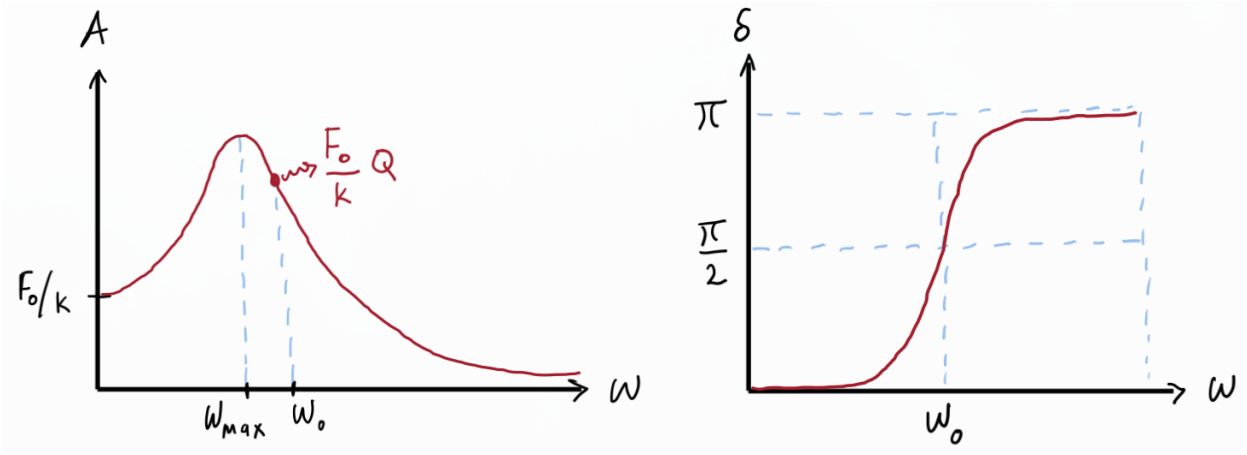


Figure 13: Amplitude and phase difference of the steady-state motion as a function of the driving frequency ω.

4.1 How to read amplitude and phase equations

(a) F_0 is the amplitude of the driving force and it is in the numerator of the amplitude equation. Therefore, increasing the driving force will increase the amplitude of the steady-state motion or vice versa.

(b) γ is the damping constant and it is in the denominator of the amplitude equation. Therefore, increasing the damping will decrease the amplitude of the steady-state motion or vice versa.

(c) What if ω, the angular frequency of the driving force, is a very small number, $\omega \ll 1$? It physically means that driving force oscillates so slow that the particle directly follows the driving force. At each instant of time, we would expect the restoring force to be approximately equal to the driving force, i.e. $kx(t) \approx F_0 \cos(\omega t)$. Therefore, the amplitude of the steady-state motion should be approximately $A \approx F_0/k$. This is exactly what we get from the amplitude equation in this limit:

$$A \approx \frac{F_0/m}{\sqrt{\omega_0^2}} = \frac{F_0/m}{\omega_0^2} = \frac{F_0/m}{k/m} = \frac{F_0}{k}. \tag{109}$$

Also, since the particle is directly following the driving force, we would expect the phase difference to be approximately zero, $\delta \approx 0$. This is exactly what we get from the phase equation in this limit:

$$\delta \approx \arctan\left(\frac{0}{\omega_0^2}\right) = \arctan(0) = 0. \tag{110}$$

(d) What if ω, the angular frequency of the driving force, is equal to the natural angular frequency of the system, $\omega = \omega_0$?

$$A = \frac{F_0/m}{\sqrt{(2\gamma\omega_0)^2}} = \frac{F_0/m}{2\gamma\omega_0}. \tag{111}$$

Remember that the quality factor is defined as $Q = \omega_0/2\gamma$. Therefore, we can write the amplitude at resonance as

$$A_{\text{res}} = \frac{F_0/m}{2\gamma\omega_0} = \frac{F_0/m}{\omega_0^2}\frac{\omega_0}{2\gamma} = \frac{F_0/k}{2\gamma/\omega_0} = \frac{F_0/k}{1/Q} = Q\frac{F_0}{k}. \tag{112}$$

Therefore, the amplitude at resonance is Q times larger than the amplitude at low frequencies. This is a very important result, and it shows that the quality factor is a measure of how much the amplitude of the steady-state motion is amplified at resonance. The phase difference at resonance is

$$\delta = \arctan\left(\frac{2\gamma\omega_0}{\omega_0^2 - \omega_0^2}\right) = \arctan(\infty) = \frac{\pi}{2}. \tag{113}$$

(e) What if ω, the angular frequency of the driving force, is a very large number, $\omega \gg 1$? It physically means that driving force oscillates so fast that the particle cannot follow the driving force. At each instant of time, we would expect the particle to be approximately at rest, i.e. $x(t) \approx 0$. Therefore, the amplitude of the steady-state motion should be approximately $A \approx 0$. This is exactly what we get from the amplitude equation in this limit:

$$A \approx \frac{F_0/m}{\sqrt{\omega^4}} = \frac{F_0/m}{\omega^2} \approx 0. \tag{114}$$

In Eq. 108, as we increase ω, the parameter of the arctan function goes to zero from the left. Therefore, the phase difference approaches π from below, $\delta \to \pi^-$. This means that the particle is approximately at rest, and the driving force is approximately equal to the restoring force, but in the opposite direction, $F_0\cos(\omega t) \approx -kx(t)$.

(f) The maximum of the amplitude occurs not at the resonance frequency ω_0, but at a slightly lower frequency ω_{max}:

$$\frac{dA}{d\omega} = \frac{F_0}{m}\left[-\frac{1}{2}\left(4\gamma^2\omega^2 + (\omega^2 - \omega_0^2)^2\right)\right]^{-3/2}\left(8\gamma^2\omega + 4\omega(\omega^2 - \omega_0^2)\right) = 0. \tag{115}$$

$$8\gamma^2\omega + 4\omega(\omega^2 - \omega_0^2) = 0 \implies 2\gamma^2 + \omega^2 - \omega_0^2 = 0 \implies \omega_{\text{max}} = \sqrt{\omega_0^2 - 2\gamma^2}. \tag{116}$$

Therefore, $\omega_{\text{max}} < \omega_0$. We can also write ω_{max} in terms of the quality factor Q:

$$\omega_{\text{max}} = \sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1 - \frac{1}{2Q^2}}. \tag{117}$$

If we plug ω_{max} into the amplitude equation, we can find the maximum amplitude:

$$A_{\text{max}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_{\text{max}}^2)^2 + (2\gamma\omega_{\text{max}})^2}} = \frac{F_0/m}{\sqrt{(2\gamma^2)^2 + (2\gamma\sqrt{\omega_0^2 - 2\gamma^2})^2}} = \frac{F_0/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}}. \tag{118}$$

In terms of the quality factor Q, we can write

$$A_{\max} = \frac{F_0/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} = \frac{F_0/m}{2\gamma\omega_0\sqrt{1-\frac{1}{4Q^2}}} = \frac{F_0/k}{\sqrt{1-\frac{1}{4Q^2}}}\frac{1}{2\gamma/\omega_0} = \frac{F_0/k}{\sqrt{1-\frac{1}{4Q^2}}}\frac{1}{1/Q} = \frac{F_0}{k}\frac{Q}{\sqrt{1-\frac{1}{4Q^2}}}. \tag{119}$$

For example, for a very low quality factor like $Q = 5$:

$$\omega_{\max} = \sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1-\frac{1}{2Q^2}} = \omega_0\sqrt{1-\frac{1}{50}} \approx 0.99\omega_0. \tag{120}$$

Therefore, even for this very low quality factor, the maximum amplitude occurs at a frequency that is very close to the natural frequency of the system. For a higher quality factors the difference between $\omega_{\max}$ and ω_0 becomes even smaller. When we compare the maximum amplitude $A_{\max}$ with the amplitude at resonance A_{res}, we can see that they are very close to each other for a low quality factor like $Q = 5$:

$$A_{\max} = \frac{F_0}{k}\frac{Q}{\sqrt{1-\frac{1}{4Q^2}}} = \frac{F_0}{k}\frac{5}{\sqrt{1-\frac{1}{100}}} \approx 5.025\frac{F_0}{k}, \quad A_{\text{res}} = Q\frac{F_0}{k} = 5\frac{F_0}{k}. \tag{121}$$

In Fig. 14, we plot the amplitude response of a driven damped harmonic oscillator for different quality factors. As we can see, the maximum amplitude occurs at a frequency that is very close to the natural frequency of the system, and the difference between $\omega_{\max}$ and ω_0 becomes smaller as the quality factor increases.

Even if the maximum amplitude occurs at a frequency $\omega_{\max}$, the amplitude at resonance A_{res} is still a very good approximation for the maximum amplitude $A_{\max}$, especially for high quality factors. For this reason, by definiton, the natural oscillation frequency ω_0 is also called the *resonance frequency* of the system, and the amplitude at resonance A_{res} is also called the *resonance amplitude* of the system.

5 Worked Examples

5.1 Work, Potential Energy, and Energy Conservation in a Spring

We are given a spring-mass system with a spring constant k and a mass m. By using the definition of work,

(a) Calculate the potential energy of the spring at a point x_0 with respect to its equilibrium position, where $U(0) = 0$.

(b) Calculate the work done by an external agent to bring the mass quasistatically from a point x_0 to the equilibrium position of the spring.

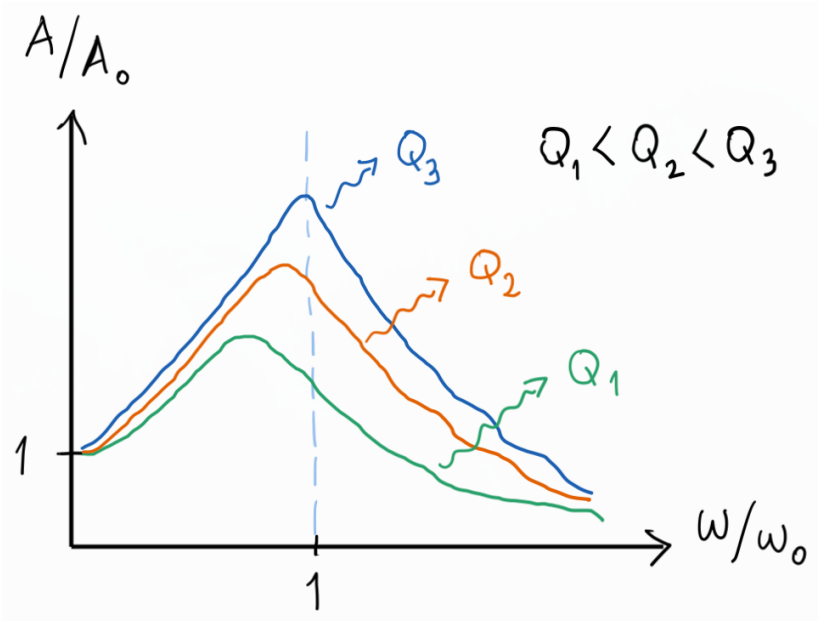


Figure 14: Amplitude response of a driven damped harmonic oscillator for different quality factors. The maximum amplitude occurs at a frequency that is very close to the natural frequency of the system, and the difference between $\omega_{\max}$ and ω_0 becomes smaller as the quality factor increases.

(c) If the force applied by an external agent is twice as large as the spring force in magnitude but opposite in direction at every position, calculate the work done by the external agent to bring the mass from a point x_1 to x_2, where $x_1 < x_2$.

Solution:

According to Hooke's law, the force exerted by the spring is

$$F_s(x) = -kx. \tag{122}$$

When the mass is moved quasistatically, the external force balances the spring force:

$$F_{\text{ext}}(x) = -F_s(x) = kx. \tag{123}$$

(a) The work done by an external agent in moving the mass quasistatically from the equilibrium position to x_0 is stored as potential energy in the spring, since the kinetic energy of the mass does not change. Therefore,

$$U(x_0) = W_{\text{ext}} = \int_0^{x_0} F_{\text{ext}}(x)\,dx = \int_0^{x_0} kx\,dx = \boxed{\frac{1}{2}kx_0^2}. \tag{124}$$

(b) When the mass is moved quasistatically from x_0 to the equilibrium position, the external force remains equal and opposite to the spring force. Thus,

$$W_{\text{ext}} = \int_{x_0}^{0} F_{\text{ext}}(x)\,dx = \int_{x_0}^{0} kx\,dx = \boxed{-\frac{1}{2}kx_0^2}. \tag{125}$$

The external agent does negative work because its force opposes the displacement toward equilibrium. The spring does an equal amount of positive work:

$$W_s = -W_{\text{ext}} = \frac{1}{2}kx_0^2. \tag{126}$$

Consequently, the decrease in potential energy is equal to the work done by the spring.

(c) The external force is twice as large as the spring force in magnitude and acts in the opposite direction:

$$F_{\text{ext}}(x) = -2F_s(x) = 2kx. \tag{127}$$

Therefore, the work done by the external agent from x_1 to x_2 is

$$W_{\text{ext}} = \int_{x_1}^{x_2} F_{\text{ext}}(x)\,dx = \int_{x_1}^{x_2} 2kx\,dx = \boxed{k(x_2^2 - x_1^2)}. \tag{128}$$

Since the external force is not balanced by the spring force, the kinetic energy of the mass can change. Assuming no other forces do work, the work-energy theorem gives

$$\Delta K = W_{\text{ext}} + W_s, \tag{129}$$

where

$$W_s = \int_{x_1}^{x_2} (-kx)\,dx = -\frac{1}{2}k(x_2^2 - x_1^2). \tag{130}$$

Hence,

$$\boxed{\Delta K = \frac{1}{2}k(x_2^2 - x_1^2)}. \tag{131}$$

Thus, half of the external work changes the spring's potential energy, while the remaining half changes the kinetic energy of the mass:

$$W_{\text{ext}} = \Delta U + \Delta K. \tag{132}$$

5.2 Simple (mathematical) pendulum

A mass m is suspended from a fixed point by a light, inextensible string of length L as shown in Figure 15. The mass is displaced to an angle θ_0 from the vertical and released from rest.

(a) Write down the equation of motion for the pendulum.

(b) Determine the period of oscillation for small angles.

(c) How the angular displacement, angular velocity and angular acceleration of the pendulum vary with time for small angles?

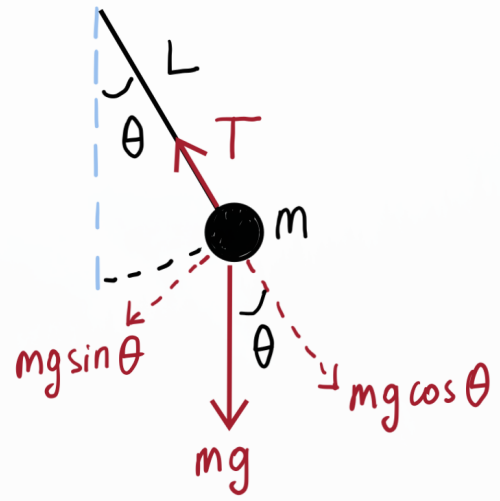


Figure 15: Simple pendulum. A mass m is suspended from a fixed point by a light, inextensible string of length L. The mass is displaced to an angle θ_0 from the vertical and released from rest.

Solution:

(a) Since the particle always moves along a circular arc, thinking in terms of the rotational form of Newton's second law is more convenient. The torque about the pivot point is given by

$$\tau = -mgL\sin\theta, \tag{133}$$

where the negative sign indicates that the torque acts to restore the pendulum to its equilibrium position. The moment of inertia of the mass about the pivot point is

$$I = mL^2. \tag{134}$$

Therefore, the equation of motion for the pendulum is

$$I\ddot{\theta} = \tau \implies mL^2\ddot{\theta} = -mgL\sin\theta \implies \ddot{\theta} + \frac{g}{L}\sin\theta = 0. \tag{135}$$

(b) For small angles, we can use the small-angle approximation $\sin\theta \approx \theta$. Therefore, the equation of motion becomes

$$\ddot{\theta} + \frac{g}{L}\theta = 0. \tag{136}$$

This is a simple harmonic oscillator equation with angular frequency

$$\omega_0 = \sqrt{\frac{g}{L}}. \tag{137}$$

The period of oscillation is given by

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{L}{g}}. \tag{138}$$

(c) Since Eq. (136) is a simple oscillator equation, we already know the general solution:

$$\theta(t) = \theta_0 \cos(\omega_0 t), \tag{139}$$

where θ_0 is the initial angular displacement. The angular velocity as a function of time is given by

$$\omega(t) = \dot{\theta}(t) = -\theta_0 \omega_0 \sin(\omega_0 t). \tag{140}$$

The angular acceleration as a function of time is given by

$$\alpha(t) = \dot{\omega}(t) = -\theta_0 \omega_0^2 \cos(\omega_0 t). \tag{141}$$

5.3 Crossing the Equilibrium Position in Critically Damped Motion

Consider a critically damped oscillator with damping constant $\gamma = 1\,\mathrm{s}^{-1}$. Initially, the particle is at $x_0 = 1\,\mathrm{m}$ and has velocity $v_0 = -2\,\mathrm{m/s}$.

(a) Determine the position of the particle as a function of time.

(b) Find the time at which the particle crosses the equilibrium position.

(c) Determine the maximum displacement of the particle on the opposite side of equilibrium and explain why the motion is not oscillatory.

Solution:

(a) The general solution for critically damped motion is

$$x(t) = [x_0 + (v_0 + \gamma x_0)t]\, e^{-\gamma t}. \tag{142}$$

Substituting the given initial conditions, we obtain

$$\boxed{x(t) = (1 - t)e^{-t},} \tag{143}$$

where t is measured in seconds and x in meters.

(b) The particle crosses the equilibrium position when $x(t) = 0$. Therefore,

$$(1 - t)e^{-t} = 0. \tag{144}$$

Since the exponential function never vanishes, we must have

$$1 - t = 0. \tag{145}$$

Thus, the particle crosses the equilibrium position at

$$\boxed{t = 1\,\mathrm{s}.} \tag{146}$$

(c) To determine the maximum displacement on the opposite side of equilibrium, we find when the velocity vanishes. Differentiating the position with respect to time gives

$$v(t) = \dot{x}(t) = (t-2)e^{-t}. \tag{147}$$

Setting $v(t) = 0$, we obtain

$$(t-2)e^{-t} = 0, \tag{148}$$

which gives

$$t = 2\,\mathrm{s}. \tag{149}$$

Substituting this into the position function, we find

$$x(2) = (1-2)e^{-2} = -e^{-2}\,\mathrm{m}. \tag{150}$$

Therefore, the particle reaches its maximum displacement on the opposite side of equilibrium at

$$\boxed{x_{\min} = -e^{-2}\,\mathrm{m} \approx -0.135\,\mathrm{m}.} \tag{151}$$

The particle initially moves toward equilibrium, crosses it at $t = 1\,\mathrm{s}$, and continues into the negative region. At $t = 2\,\mathrm{s}$, it momentarily stops and reverses direction, approaching equilibrium from below as $t \to \infty$.

Although the particle crosses the equilibrium position once, the motion is not oscillatory because it never crosses equilibrium again. This illustrates that critical damping does not necessarily prevent a particle from crossing equilibrium; rather, it prevents repeated oscillations about equilibrium.

5.4 Kinetic energy resonance in a driven damped harmonic oscillator

At which driving frequency ω does the average kinetic energy of a driven damped harmonic oscillator reach its maximum?

Solution:

The steady-state displacement of a driven damped harmonic oscillator is

$$x(t) = A\cos(\omega t + \phi), \tag{152}$$

where ω is the driving frequency, A is the displacement amplitude, and ϕ is the phase. The corresponding velocity is

$$v(t) = \dot{x}(t) = -\omega A\sin(\omega t + \phi). \tag{153}$$

Therefore, the instantaneous kinetic energy is

$$T(t) = \frac{1}{2}mv^2(t) = \frac{1}{2}m\omega^2 A^2\sin^2(\omega t + \phi). \tag{154}$$

The average kinetic energy over one complete period $P = 2\pi/\omega$ is defined as

$$\langle T \rangle = \frac{1}{P} \int_0^P T(t)\, dt. \tag{155}$$

Substituting the instantaneous kinetic energy gives

$$\langle T \rangle = \frac{m\omega^2 A^2}{2P} \int_0^P \sin^2(\omega t + \phi)\, dt \tag{156}$$

$$= \frac{m\omega^2 A^2}{4P} \int_0^P [1 - \cos(2\omega t + 2\phi)]\, dt \tag{157}$$

$$= \frac{1}{4} m\omega^2 A^2, \tag{158}$$

where we used the identity $\sin^2\theta = (1 - \cos 2\theta)/2$ and the fact that the cosine term averages to zero over a complete period. The steady-state displacement amplitude is

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \tag{159}$$

where F_0 is the amplitude of the driving force, ω_0 is the natural frequency, and $\gamma > 0$ is the damping constant. Substituting this expression into the average kinetic energy, we obtain

$$\langle T \rangle = \frac{F_0^2}{4m} \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}. \tag{160}$$

For $\omega > 0$, dividing the numerator and denominator by ω^2 gives

$$\langle T \rangle = \frac{F_0^2}{4m} \frac{1}{(\omega_0^2/\omega - \omega)^2 + 4\gamma^2}. \tag{161}$$

The average kinetic energy is maximized when the denominator is minimized. Since the squared term is nonnegative, its minimum occurs when

$$\frac{\omega_0^2}{\omega} - \omega = 0. \tag{162}$$

Thus, the driving frequency at which the average kinetic energy reaches its maximum is

$$\boxed{\omega_{\text{res}} = \omega_0.} \tag{163}$$

Conclusion: The kinetic energy resonance occurs at the natural frequency ω_0, independently of the damping constant γ. This differs from displacement resonance, which occurs at

$$\omega_{\text{res}}^{(x)} = \sqrt{\omega_0^2 - 2\gamma^2}, \tag{164}$$

provided $\gamma < \omega_0/\sqrt{2}$. That's the reason why we call the natural frequency ω_0 the resonance frequency of the system.

5.5 Amplitude damping

The amplitude of a lightly damped oscillator decreases by 5% each cycle. What percentage of mechanical energy is lost per cycle?

Solution: The mechanical energy of a harmonic oscillator is proportional to the square of its amplitude. If the amplitude decreases by 5% each cycle, the new amplitude after one cycle is

$$A_{\text{new}} = 0.95 A_{\text{old}}. \tag{165}$$

The mechanical energy after one cycle is

$$E_{\text{new}} \propto A_{\text{new}}^2 = (0.95 A_{\text{old}})^2 = 0.9025 A_{\text{old}}^2. \tag{166}$$

Therefore, the percentage of mechanical energy lost per cycle is

$$\text{Percentage lost} = \left(1 - \frac{E_{\text{new}}}{E_{\text{old}}}\right) \times 100\% = (1 - 0.9025) \times 100\% = 9.75\%. \tag{167}$$

5.6 Coupled Oscillators

Consider two masses m_1 and m_2 connected by a spring with spring constant k, moving along a frictionless horizontal surface. Let x_1 and x_2 denote their displacements from their respective equilibrium positions. Show that the relative displacement $x = x_2 - x_1$ undergoes simple harmonic motion, and determine its angular frequency.

Solution:

By Newton's second law, the equations of motion are

$$m_1 \ddot{x}_1 = k(x_2 - x_1), \tag{168}$$

$$m_2 \ddot{x}_2 = -k(x_2 - x_1). \tag{169}$$

Introducing the relative displacement $x = x_2 - x_1$, we obtain

$$\ddot{x} = \ddot{x}_2 - \ddot{x}_1 = -k\left(\frac{1}{m_1} + \frac{1}{m_2}\right) x. \tag{170}$$

Rearranging,

$$\ddot{x} + k\left(\frac{1}{m_1} + \frac{1}{m_2}\right) x = 0. \tag{171}$$

Comparing with the equation of a simple harmonic oscillator, $\ddot{x} + \omega^2 x = 0$, we find

$$\boxed{\omega = \sqrt{k\left(\frac{1}{m_1} + \frac{1}{m_2}\right)}}. \tag{172}$$

Thus, although both masses are free to move, their relative displacement undergoes simple harmonic motion with angular frequency ω. We define the reduced mass μ as

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} \implies \mu = \frac{m_1 m_2}{m_1 + m_2}. \tag{173}$$

Therefore, the angular frequency can also be expressed as

$$\omega = \sqrt{\frac{k}{\mu}}. \tag{174}$$

5.7 Determining the damping constant from amplitude decay

The amplitude of a damped harmonic oscillator reduces to the half of its initial value in 1 second. Find the damping constant γ.

Solution: The amplitude of a damped harmonic oscillator is given by

$$A(t) = A_0 e^{-\gamma t}. \tag{175}$$

If the amplitude reduces to half its initial value in 1 second, then

$$\frac{1}{2} A_0 = A_0 e^{-\gamma \cdot 1}. \tag{176}$$

Simplifying,

$$\frac{1}{2} = e^{-\gamma}. \tag{177}$$

Taking the natural logarithm of both sides,

$$\ln\left(\frac{1}{2}\right) = -\gamma. \tag{178}$$

Therefore,

$$\boxed{\gamma = \ln(2)}. \tag{179}$$

5.8 Block

A block of mass m is sliding on a horizontal surface. At time $t = 0$, its velocity is v_0. There is a spring with spring constant k attached to a wall. The block collides with the spring and compresses it.

(a) Determine the maximum compression of the spring.

(b) Determine the time it takes for the block to reach maximum compression.